

Abstract

Floods typically occur due to ice jams in the winter or extended periods of precipitation in the spring and summer seasons. An increase in the rate of water discharge in the river coincides with a flood event. This research combines the time series decomposition and the time series regression model for the flood prediction in Mohawk River at Schenectady, New York. The time series decomposition has been applied to separate the different frequencies in hydrogeological and climatic data. The time series data have been decomposed into the long-term, seasonal-term, and short-term components using the Kolmogorov-Zurbenko filter. For the application of the time series regression model, we determine the lags of the hydrogeological and climatic variables that provide the maximum performance for the model. The lags applied in the predictor variables of the model have been used for the physical interpretation of the model to strengthen the relationship between the water discharge and the climatic and hydrogeological variables. The overall model accuracy has been increased up to 73%. The results show that using the lags of the variables in the time regression model, and the forecasting accuracy has been increased compared to the raw data by two times.

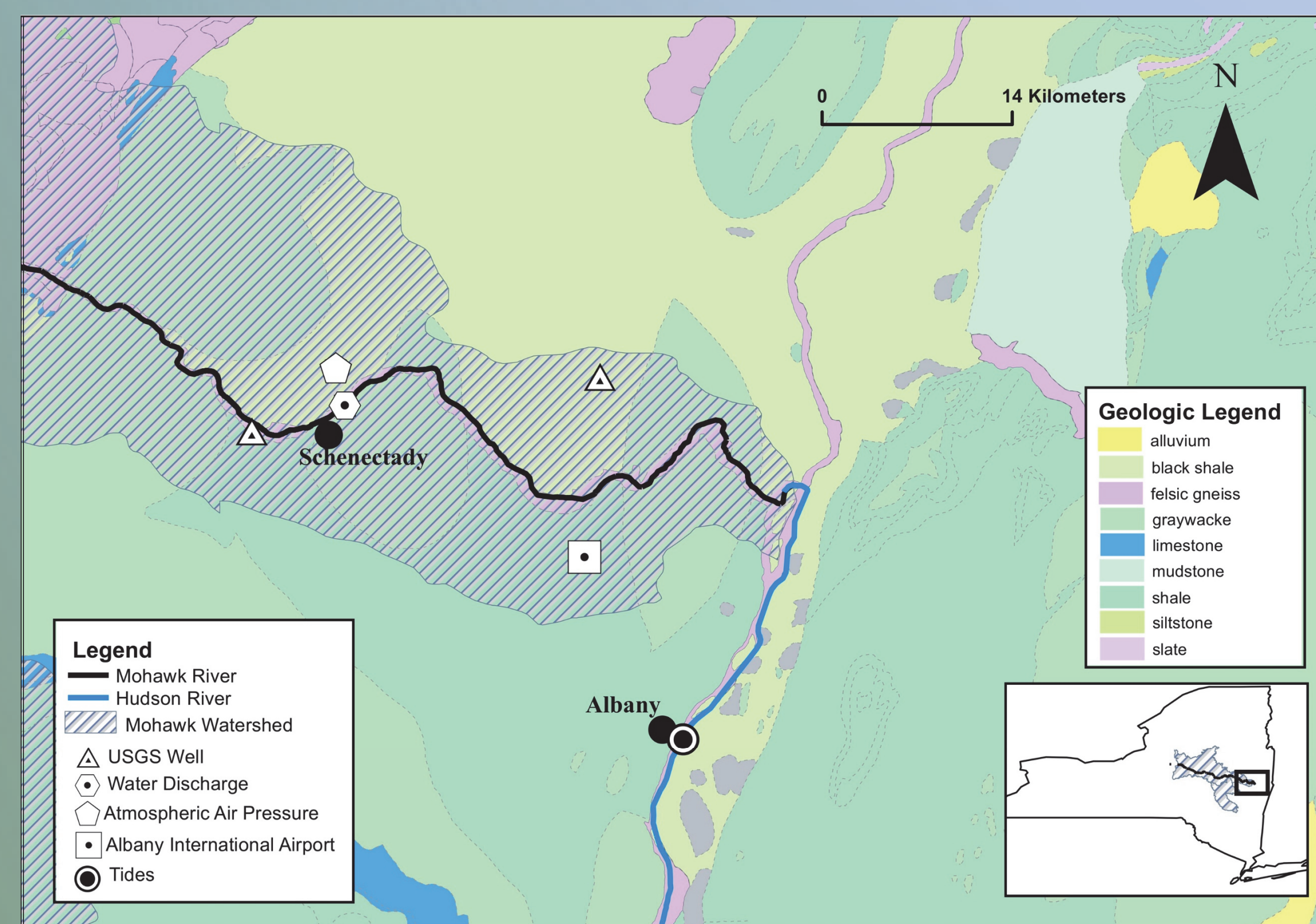


Figure 1: A map of Upstate New York made in ArcMap showing the Mohawk watershed, the Mohawk River and the location of Schenectady. All points indicate the location of the climatic and hydrogeological data that are derived from the monitoring stations. Geological data were derived from New York State.

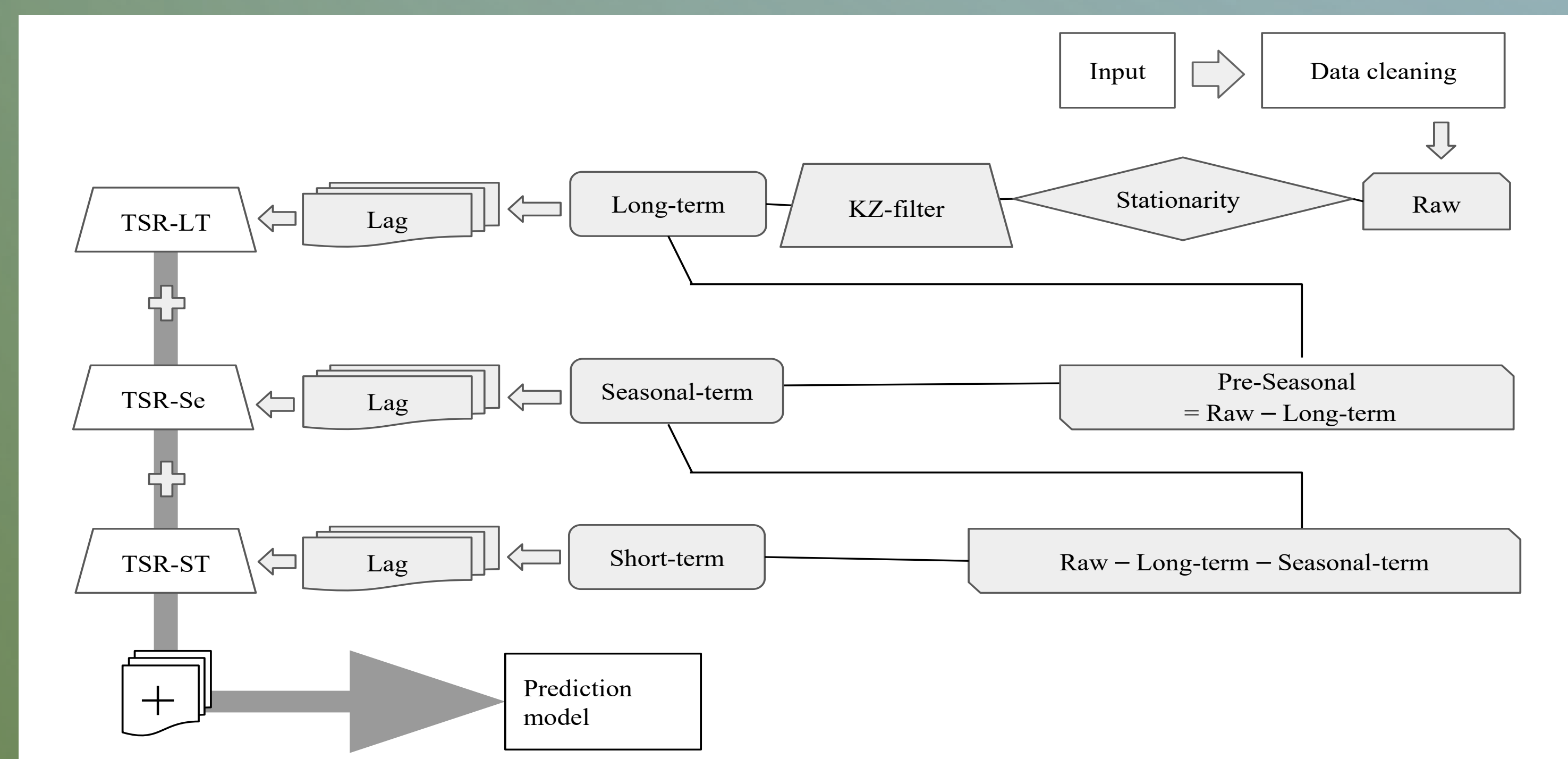


Figure 2: A flowchart displaying the steps of the methodology which includes the decomposition of the time series using the Kolmogorov-Zurbenko (KZ) filter, the lag operator, and the time series regression (TSR) model applied for each component (LT; long term, Se; seasonal term, ST; short term). The multiple linear regression model (MLR) follows a similar flow chart excluding the lag operator step.

Methods

- The data collection of the monitoring stations and preprocessing was performed in KNIME [1, 2]. A log transformation of the variables was done to stabilize the variance of the data for the series to be stationary [3]. The extracted raw data were retrieved from their original URL sources, and 15-min, hourly, and daily data were obtained for the specific location (Figure 2).
- All the data were converted to a daily time series considering the maximum value per day. The TSR model has been applied in the decomposed data to predict the water discharge of each component in the long, seasonal, and short-term component [3-5].
- The correlation coefficient has been calculated between the water discharge and the lagged predictor variables to estimate the appropriate lags in the TSR model.
- The MLR and TSR models were performed and the results of the coefficient of determination, R^2 , combined with the contribution of the variance of each component were compared to evaluate the overall performance of the TSR model (Table 1).

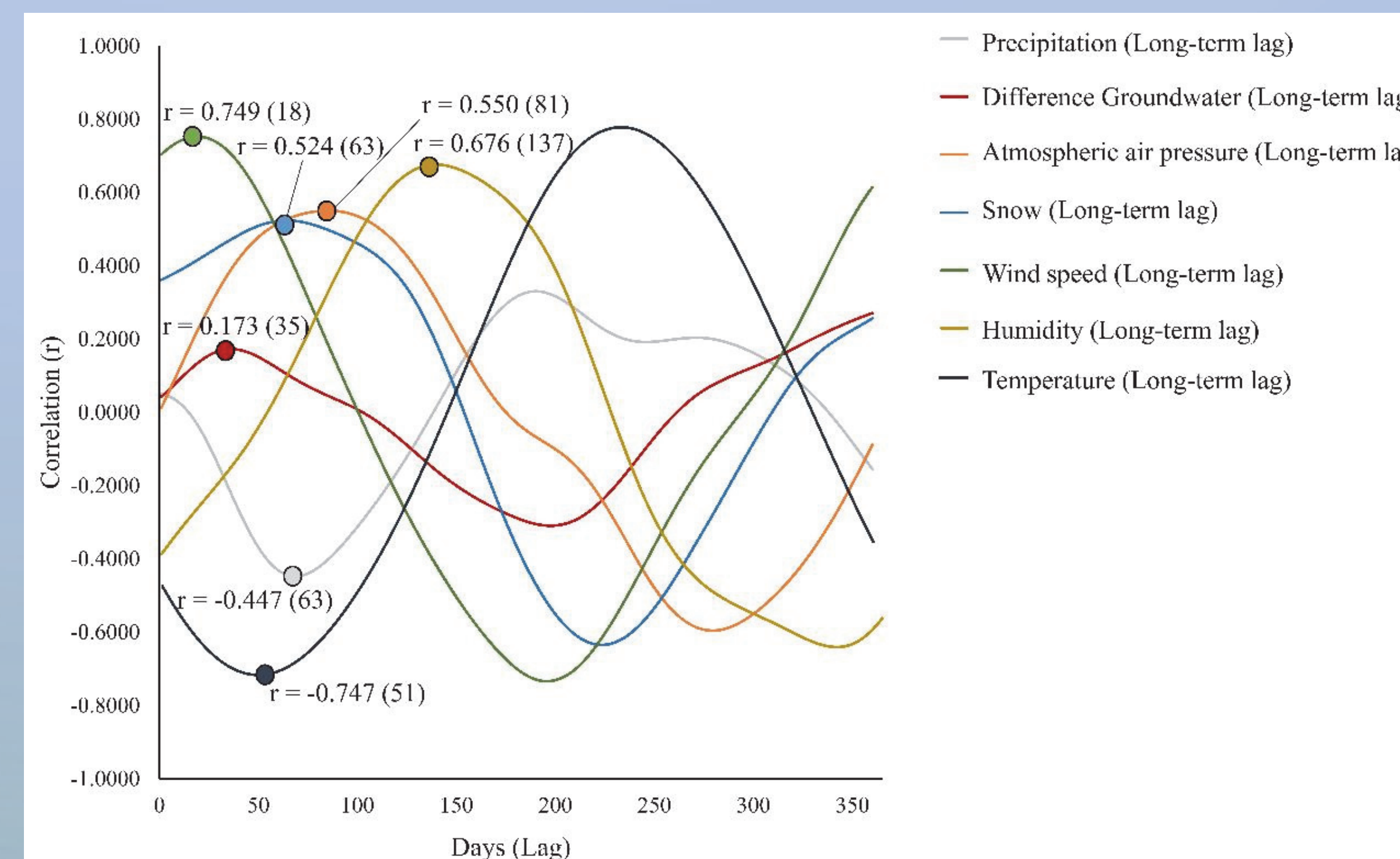


Figure 3: Determination of the strongest correlation (positive or negative) between the water discharge and the lagged hydrogeological and climatic variables for the long-term component. The strongest correlation between the water discharge and the predictor variables is equal to -1 or 1, while the weakest correlation occurs when the value is 0. The shaded points represent the strongest correlation value.

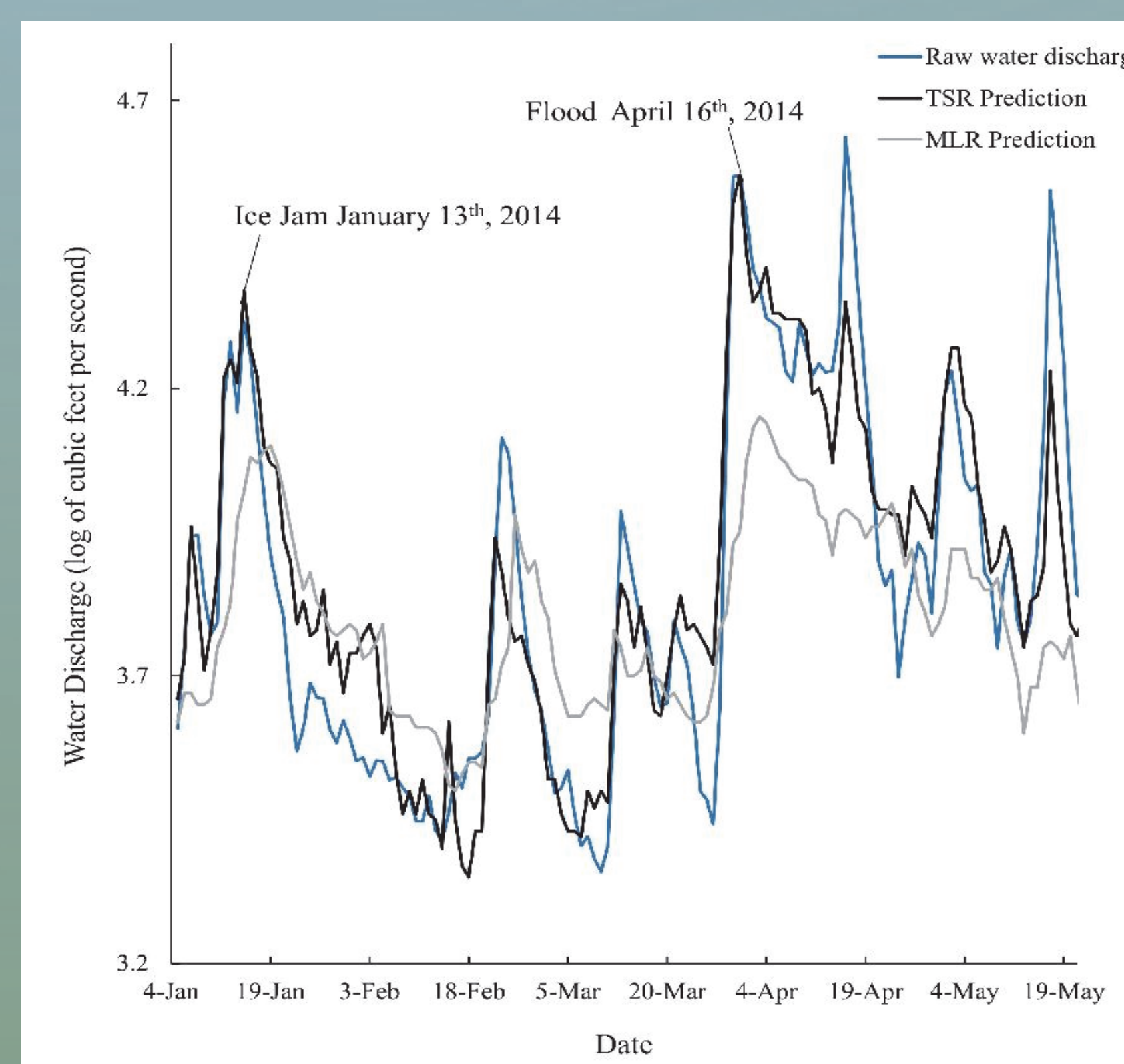


Figure 4: The raw water discharge data (blue line), the water discharge data derived from the TSR model (black line), and the water discharge data derived from the multiple linear regression (MLR) model (grey line). The water discharge data is the logarithmic transformation of the cubic meters per second values measured from 4 January 2014 to 19 May 2014.

Results

Table 1: Comparison of the coefficient determination (R^2) for the MLR and TSR model

	LT	Se	ST
R^2 (MLR)	0.74	0.50	0.29
R^2 (TSR)	0.93	0.58	0.41

Table 2: The first row describes the coefficient of determination (R^2) of the TSR model for the long, seasonal, and short-term component of the water discharge data. The second row describes the percent of the variance contributed in each component. The third row is the percent of explanation described for each component, and the fourth row is the total explanation of the TSR model.

	Raw Data	LT	Se	ST
Coefficient Determination (R^2)	0.49	0.93	0.58	0.41
Variance %		59.02	9.02	31.96
Explanation %		54.89	5.23	13.10
Total Explanation %				73.22

Discussion

- The TSR model provides a higher forecasting performance compared to the MLR model (figure 4).
- The TSR is a dynamic model compared to the MLR model and provides the advantage of considering past values for the hydrogeological and climatic variable.
- Using the lag operator in the time series regression model, we can reveal the true relationship between the water discharge and the hydrological and climatic variables.

Conclusion

The separation of the data into three different time series components named as the long, seasonal, and short-term increases the accuracy of the model compared to the raw data. The TSR model outperforms the MLR model for predicting an increase in water discharge due to flood events and ice jams. Lag values may provide the threshold values and subsequent dates at which the independent variables may contribute to water discharge rise and pose a high risk of a flood which could be triggered in case of an oncoming storm or ice jam occurrence.

References

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